Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: **BTech (ECE) VII**
Name of the Paper: **Neural Network and Machine Learning**
Paper Code: **TOE 701**
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. What are the key differences between biological neural networks and artificial neural networks (ANNs)? CO1

OR

b. Explain the primary applications of ANNs. CO1

Q2. (10 Marks)

a. Describe the structure and function of a computational model of a neuron. CO1

OR

b. What are the different structures of neural networks, and how do they affect performance? CO1

Q3. (10 Marks)

a. What is polynomial curve fitting, and how does it differ from linear regression? CO1

OR

b. What are linear basis function models, and how are they used in machine learning? CO1

Q4. (10 Marks)

a. What is a perceptron, and how is it used for pattern classification? CO2

OR

b. Describe the structure of a multilayer feedforward neural network. CO2

Q5. (10 Marks)

a. Explain how pattern classification is performed using MLFNNs. CO2

OR

b. What is the importance of the error function in training neural networks? CO2